

Addition of two numbers in base p

Let A and B be two integers represented in base p .

$A = (a_m a_{m-1} \dots a_1 a_0)_{(p)}$, with $m+1$ digits and

$B = (b_n b_{n-1} \dots b_1 b_0)_{(p)}$, with $n+1$ digits.

We calculate $A + B = C = (c_k c_{k-1} \dots c_1 c_0)_{(p)}$, with $k+1$ digits.

The addition of two numbers is performed starting with the units digit (index 0), from right to left, the process being repetitive with a number of $\max(m, n) + 1$ iterations.

In each iteration i , the digits (after they have been converted to decimal) of the same index (i) from the two numbers and the carry digit (0 or 1) from the previous iteration ($i-1$) are added.

The resulting *sum* provides 2 digits:

- the **carry digit** used in the next iteration is the quotient obtained by dividing *sum* by p ;
- the **positionally corresponding digit (index i)** in number C is the remainder obtained by dividing *sum* by p ;

!!! All the operations presented above are performed in decimal.

An initial carry digit, $t_0 = 0$, is considered. The number with less digits is filled with 0 digits (insignificant digits) to the left, to have $\max(m, n) + 1$ digits in both numbers. The addition operation is described by the following mathematical formulas:

$$c'_i = (a'_i + b'_i + t_i) - \left\lfloor \frac{a'_i + b'_i + t_i}{p} \right\rfloor * p, i = 0, 1, \dots, \max(m, n)$$

$$t_{i+1} = \left\lfloor \frac{a'_i + b'_i + t_i}{p} \right\rfloor, i = 1, \dots, \max(m, n)$$

- where: $a'_i = (a_i)_{(p)}$, $b'_i = (b_i)_{(p)}$ (the digits of both numbers are converted to decimal)

- now all the digits in the formulas are decimal digits and the calculations are performed in decimal

c'_i = the remainder obtained by dividing $a'_i + b'_i + t_i$ by p

$(c_i)_{(p)} = c'_i$ (the decimal digit c'_i is converted in a digit of base p)

t_{i+1} = the quotient obtained by dividing $a'_i + b'_i + t_i$ by p

If $t_{\max(m, n) + 1} = 0$ then $k = \max(m, n)$,

else $k = \max(m, n) + 1$ and $c_k = t_k$ (is 1, being the last carry digit)

Example 1: $43012_{(5)} + 3243_{(5)} = ?_{(5)}$

$ \begin{array}{r} 110110 \\ 43012_{(5)} + \\ 03243_{(5)} \\ \hline 101310_{(5)} \end{array} $	$ \begin{array}{l} m=4, n=3 \\ i=0: \quad t_0 = 0 \\ 2_{(5)} + 3_{(5)} + 0_{(5)} = 2 + 3 + 0 = 5 \\ t_1 = [5/5] = 1, \quad c'_0 = 5 - 1 * 5 = 0 \\ c_0 = 0_{(5)} \\ i=1: \quad t_1 = 1 \\ 1_{(5)} + 4_{(5)} + 1_{(5)} = 1 + 4 + 1 = 6 \end{array} $	$ \begin{array}{l} i=2: \quad t_2 = 1 \\ 0_{(5)} + 2_{(5)} + 1_{(5)} = 0 + 2 + 1 = 3 \\ t_3 = [3/5] = 0, \quad c'_2 = 3 - 0 * 5 = 3 \\ c_2 = 3_{(5)} \\ i=3: \quad t_3 = 0 \\ 3_{(5)} + 3_{(5)} + 0_{(5)} = 3 + 3 + 0 = 6 \end{array} $
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$$\begin{aligned}
t_2 &= \lfloor 6/5 \rfloor = 1, \quad c'_1 = 6 - 1 \cdot 5 = 1 \\
c_1 &= 1_{(5)} \\
i=4: \quad t_4 &= 1 \\
4_{(5)} + 0_{(5)} + 1_{(5)} &= 4 + 0 + 1 = 5 \\
t_5 &= \lfloor 5/5 \rfloor = 1, \quad c'_4 = 5 - 1 \cdot 5 = 0 \\
c_4 &= 0_{(5)} \\
k &= 5, \quad c_5 = t_5 = 1_{(5)} \\
43012_{(5)} + 3243_{(5)} &= 101310_{(5)}
\end{aligned}$$

Example 2: $A5F_{(16)} + 96BD_{(16)} = ?_{(16)}$

$$\begin{array}{rcl}
\begin{array}{l}
01110 \\
0A5F_{(16)} + \\
96BD_{(16)} \\
\hline
A11C_{(16)}
\end{array} & \begin{array}{l}
m=3, n=2 \\
i=0: \quad t_0 = 0 \\
F_{(16)} + D_{(16)} + 0_{(16)} = 15 + 13 + 0 = 28 \\
t_1 = \lfloor 28/16 \rfloor = 1, \quad c'_0 = 28 - 1 \cdot 16 = 12 \\
c_0 = C_{(16)} \\
i=1: \quad t_1 = 1 \\
5_{(16)} + B_{(16)} + 1_{(16)} = 5 + 11 + 1 = 17 \\
t_2 = \lfloor 17/16 \rfloor = 1, \quad c'_1 = 17 - 16 \cdot 1 = 1 \\
c_1 = 1_{(16)} \\
k = 3, \quad t_4 = 0 \\
A5F_{(16)} + 96BD_{(16)} = A11C_{(16)}
\end{array} & \begin{array}{l}
i=2: \quad t_2 = 1 \\
A_{(16)} + 6_{(16)} + 1_{(16)} = 10 + 6 + 1 = 17 \\
t_3 = \lfloor 17/16 \rfloor = 1, \quad c'_2 = 17 - 16 \cdot 1 = 1 \\
c_2 = 1_{(16)} \\
i=3: \quad t_3 = 0 \\
0_{(16)} + 9_{(16)} + 1_{(16)} = 0 + 9 + 1 = 10 \\
t_4 = \lfloor 10/16 \rfloor = 0, \quad c'_3 = 10 - 0 \cdot 16 = 10 \\
c_3 = A_{(16)}
\end{array}
\end{array}$$

Example 3: $1110101101_{(2)} + 110110011_{(2)} = ?_{(2)}$

$$\begin{array}{r}
1110111110 \\
1110101101_{(2)} + \\
0110110011_{(2)} \\
\hline
10101100000_{(2)}
\end{array}$$

Multiplication of a number by a digit in base p

Let A be a integer and b a digit, both written in base p :

$$A = (a_m a_{m-1} \dots a_1 a_0)_{(p)}, \text{ with } m+1 \text{ digits and } b_{(p)}.$$

We calculate the product: $A \cdot b = C = (c_k c_{k-1} \dots c_1 c_0)_{(p)}$, with $k+1$ digits.

The multiplication of a number by a digit is performed starting with the units digit (index 0), from right to left, the process being repetitive with a number of $m+1$ iterations.

At each iteration i , the current digit in A is multiplied by b , after they have been converted to decimal, and then the carry digit from the previous iteration ($i-1$) is added. The calculated *value* provides 2 digits:

- the **carry digit** used in the next iteration is the quotient obtained by dividing *value* by p ;
- the **positionally corresponding digit** (index i) in C is the remainder obtained by dividing *value* by p .

!!! All the operations presented above are performed in decimal.

An initial carry digit, $t_0 = 0$, is considered. The multiplication operation is described by the following mathematical formulas:

$$k = m$$

$$c'_i = (a'_i * b' + t_i) - \left\lfloor \frac{a'_i * b' + t_i}{p} \right\rfloor * p, i = \overline{0, k}$$

$$t_{i+1} = \left\lfloor \frac{a'_i * b' + t_i}{p} \right\rfloor, i = \overline{0, k}$$

If $t_{m+1} \neq 0$ then $k = m + 1$.

- where: $a'_i = (a_i)_{(p)}$, $b'_i = (b_i)_{(p)}$ (the digits of A and b are converted to decimal)
- the calculations in the above formulas are performed in base 10
- t_{i+1} = the quotient obtained by dividing $a'_i * b' + t_i$ by p ; $t_i, i = \overline{0, m}$ are decimal values
- c'_i = the remainder obtained by dividing $a'_i * b' + t_i$ by p

$(c_i)_{(p)} = c'_i$ (the decimal digit c'_i is converted to a digit of base p)

The product C has $m+2$ digits if $t_{m+1} \neq 0$,

else C has $m+1$ digits.

Example 1: $2031_{(4)} * 3_{(4)} = ?_{(4)}$

$ \begin{array}{r} 10200 \\ 2031_{(4)} * \\ \underline{3_{(4)}} \\ 12213_{(4)} \end{array} $	$ \begin{aligned} m &= 3 \\ i=0: \quad t_0 &= 0 \\ 1_{(4)} * 3_{(4)} + 0_{(4)} &= 1*3 + 0 = 3 \\ t_1 &= \lfloor 3/4 \rfloor = 0, \quad c'_0 = 3 - 0*4 = 3 \\ c_0 &= 3_{(4)} \\ i=1: \quad t_1 &= 0 \\ 3_{(4)} * 3_{(4)} + 0_{(4)} &= 3*3 + 0 = 9 \\ t_2 &= \lfloor 9/4 \rfloor = 2, \quad c'_1 = 9 - 2*4 = 1 \\ c_1 &= 1_{(4)} \\ k=4, t_4 &= 1, c_4 = 1 \\ 2031_{(4)} * 3_{(4)} &= 12213_{(4)} \end{aligned} $	$ \begin{aligned} i=2: \quad t_2 &= 2 \\ 0_{(4)} * 3_{(4)} + 2_{(4)} &= 0*3 + 2 = 2 \\ t_3 &= \lfloor 2/4 \rfloor = 0, \quad c'_2 = 2 - 0*4 = 2 \\ c_2 &= 2_{(4)} \\ i=3: \quad t_3 &= 0 \\ 2_{(4)} * 3_{(4)} + 0_{(4)} &= 2*3 + 0 = 6 \\ t_4 &= \lfloor 6/4 \rfloor = 1, \quad c'_3 = 6 - 1*4 = 2 \\ c_3 &= 2_{(4)} \end{aligned} $
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Example 2: $2B5F_{(16)} * A_{(16)} = ?_{(16)}$

$ \begin{array}{r} 17390 \\ 2B5F_{(16)} * \\ \underline{A_{(16)}} \\ 1B1B6_{(16)} \end{array} $	$ \begin{aligned} m &= 3 \\ i=0: \quad t_0 &= 0 \\ F_{(16)} * A_{(16)} + 0_{(16)} &= 15*10 + 0 = 150 \\ t_1 &= \lfloor 150/16 \rfloor = 9, \quad c'_0 = 150 - 9*16 = 6 \\ c_0 &= 6_{(16)} \\ i=1: \quad t_1 &= 9 \\ 5_{(16)} * A_{(16)} + 9_{(16)} &= 5*10 + 9 = 59 \\ t_2 &= \lfloor 59/16 \rfloor = 3, \quad c'_1 = 59 - 3*16 = 11 \\ c_1 &= B_{(16)} \\ k=4, t_4 &= 1 \\ 2B5F_{(16)} * A_{(16)} &= 1B1B6_{(16)} \end{aligned} $	$ \begin{aligned} i=2: \quad t_2 &= 3 \\ B_{(16)} * A_{(16)} + 3_{(16)} &= 11*10 + 3 = 113 \\ t_3 &= \lfloor 113/16 \rfloor = 7, \quad c'_2 = 113 - 7*16 = 1 \\ c_2 &= 1_{(16)} \\ i=3: \quad t_3 &= 7 \\ 2_{(16)} * A_{(16)} + 7_{(16)} &= 2*10 + 7 = 27 \\ t_4 &= \lfloor 27/16 \rfloor = 1, \quad c'_3 = 27 - 1*16 = 11 \\ c_3 &= B_{(16)} \end{aligned} $
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Division of a number by a digit in base p

Let A be a integer and b a digit, both written in base p :

$$A = (a_m a_{m-1} \dots a_1 a_0)_{(p)}, \text{ with } m+1 \text{ digits and } b_{(p)}.$$

We calculate C = the **quotient** obtained by dividing A by b , $C = (c_k c_{k-1} \dots c_1 c_0)_{(p)}$, with $k+1$ digits and the **remainder** $r_{(p)}$, $0 \leq r < b$

The division is performed starting with the most significant digit (index m), from left to right, the process being repetitive with a number of $m+1$ iterations.

At each iteration i , a value obtained as the sum of the current digit (index i) in A with the product of the carry digit from the previous iteration $i-1$ and the base p , after all digits have been converted to decimal. The calculated *value* provides 2 digits:

- the carry digit used in the next iteration is the remainder obtained by dividing *value* by b ;
- the positionally corresponding digit in C is the quotient obtained by dividing *value* by b ;

The **quotient** C will have $m+1$ digits and if the most significant digit (index m) is 0 then it will be removed.

The carry digit obtained in the last iteration will represent the **remainder** r .

An initial carry digit, $t_m = 0$, is considered. The division operation is formalized by the following mathematical formulas:

$$c'_i = \left\lfloor \frac{t_i * p + a'_i}{b'} \right\rfloor, i = \overline{m, 0}$$

$$t_{i-1} = (t_i * p + a'_i) - \left\lfloor \frac{t_i * p + a'_i}{b'} \right\rfloor * b', i = \overline{m, 0}$$

- where: $a'_i = (a_i)_{(p)}$, $b' = (b)_{(p)}$ (the digits of A and b are converted to decimal)
- the calculations in the above formulas are performed in base 10
- t_{i-1} = the remainder obtained by dividing $t_i * p + a'_i$ by b' ; $t_i, i = \overline{m, -1}$ are decimal values
- c_i = the quotient obtained by dividing $t_i * p + a'_i$ by b'
 $(c_i)_{(p)} = c'_i$ (the decimal digit c'_i is converted to a digit of base p)
- the **remainder** : $r_{(p)} = t_{-1}$

Example 1:

$\begin{array}{r} 0A5B_{(16)} \mid 8_{(16)} \\ \hline 25 \\ \hline 5B \\ \hline 3_{(16)} \text{ -- remainder: } r \end{array}$	$\begin{array}{l} m=2, 8_{(16)}=8 \\ i=2, t_2=0, \quad 0 * 16 + A_{(16)} = 0+10=10 \\ c'_2 = \lfloor 10/8 \rfloor = 1, \quad c_2 = 1_{(16)}, \quad t_1 = 10 - 8*1 = 2, \\ i=1, t_1=2, \quad 2 * 16 + 5_{(16)} = 2*16+5=37 \\ c'_1 = \lfloor 37/8 \rfloor = 4, \quad c_1 = 4_{(16)}, \quad t_0 = 37 - 8*4 = 5 \\ i=0, t_0=5, \quad 5 * 16 + B_{(16)} = 5*16+11=91 \\ c'_0 = \lfloor 91/8 \rfloor = 11, \quad c_0 = B_{(16)}, \quad t_{-1} = 91 - 8*11 = 3 \\ t_{-1} = 3 = 3_{(16)} = r \end{array}$
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Example 2:

$$\begin{array}{r}
 02043_{(8)} \mid 5_{(8)} \\
 \hline
 25 \\
 \hline
 14 \\
 \hline
 23 \\
 \hline
 4_{(8)} - \text{remainder: } r
 \end{array}
 \quad
 \begin{array}{l}
 m=3, 5_{(8)}=5 \\
 i=3, t_3=0, \quad 0*8+2_{(8)}=0+2=2 \\
 c'_3=[2/5]=0, \quad c_3=0_{(8)}, \quad t_2=2-5*0=2, \\
 i=2, t_2=2, \quad 2*8+0_{(8)}=2*8+0=16 \\
 c'_2=[16/5]=3, \quad c_2=3_{(8)}, \quad t_1=16-5*3=1 \\
 i=1, t_1=1, \quad 1*8+4_{(8)}=1*8+4=12 \\
 c'_1=[12/5]=2, \quad c_1=2_{(8)}, \quad t_0=12-5*2=2 \\
 i=0, t_0=2, \quad 2*8+3_{(8)}=2*8+3=19 \\
 c'_0=[19/5]=3, \quad c_0=3_{(8)}, \quad t_{-1}=19-5*3=4 \\
 t_{-1}=4=4_{(8)}=r
 \end{array}$$

Subtraction of two numbers in base p

Let A and B be two integers represented in base p .

$$A = (a_m a_{m-1} \dots a_1 a_0)_{(p)}, \text{ with } m+1 \text{ digits and}$$

$$B = (b_n b_{n-1} \dots b_1 b_0)_{(p)}, \text{ with } n+1 \text{ digits, } m \geq n$$

We calculate $A - B = C = (c_k c_{k-1} \dots c_1 c_0)_{(p)}$, with $k+1$ digits.

A is the **minuend**, B the **subtrahend** and C the **difference**.

The subtraction of the two numbers is performed starting with the units digit (index 0), from right to left. If necessary, the number B will be filled to the left with $m-n$ insignificant zeros.

Therefore, in the first iteration b_0 will be subtracted from a_0 . If $a_0 < b_0$, one unit will be borrowed from the higher order digit of A , which means (due to the positional writing of numbers) that the borrow is the value p of the base, and the expression $a_0 + p - b_0$ is calculated in decimal obtaining the digit c_0 . The borrow can be considered to be a carry digit that can take the value 0 or -1 in the case of a borrow.

In the next iterations we work similarly but we also add the carry from the previous iteration to the digit of A before subtracting the corresponding digit of B . The result has at most $m+1$ digits.

An initial carry digit $t_0 = 0$ is considered. If necessary, B will be filled to the left with $m-n$ insignificant zeros. The subtraction operation is formalized by the following mathematical formulas:

$$c'_i = \begin{cases} a'_i + t_i - b'_i & , \text{dacă } a'_i + t_i \geq b'_i, \text{ iar } t_{i+1} = 0 \\ a'_i + t_i + p - b'_i & , \text{dacă } a'_i + t_i < b'_i, \text{ iar } t_{i+1} = -1 \end{cases}, i = \overline{0, m}$$

The leftmost insignificant zero digits of C will be eliminated. The number of digits k will be calculated such as $c_k \neq 0$ and $\forall l > k, l \leq m, c_l = 0$.

- where: $a'_i = (a_i)_{(p)}$, $b'_i = (b_i)_{(p)}$ (the digits of A and B are converted to decimal)
- the calculations in the above formulas are performed in base 10
- c'_i (the decimal digit c'_i is converted to a digit of base p)

Example 1: $A11C_{(16)} - A5F_{(16)} = ?_{(16)}$ $-1 \ -1 \ -1 \ 0$ $m=3, n=2, t_0 = 0$ $A11C_{(16)} - i=0 :$ $0A5F_{(16)} \quad C_{(16)} + 0_{(16)} < F_{(16)} \Rightarrow \text{borrow one unit from position } i+1=1, \text{ which becomes } 10_{(16)} \text{ in the current position and } t_1 = -1$ $96BD_{(16)} \quad C_{(16)} + 0_{(16)} + 10_{(16)} - F_{(16)} = (\text{conversion to decimal}) = 12 + 16 - 15 = 13 = c'_0$ $c_0 = D_{(16)}$ $i=1 : t_1 = -1$ $1_{(16)} + (-1)_{(16)} < 5_{(16)} \Rightarrow \text{borrow} \Rightarrow t_2 = -1$ $1_{(16)} + (-1)_{(16)} + 10_{(16)} - 5_{(16)} = 1 - 1 + 16 - 5 = 11 = c'_1$ $c_1 = B_{(16)}$ $i=2 : t_2 = -1$ $1_{(16)} + (-1)_{(16)} < A_{(16)} \Rightarrow \text{borrow} \Rightarrow t_3 = -1$ $1_{(16)} + (-1)_{(16)} + 10_{(16)} - A_{(16)} = 1 - 1 + 16 - 10 = 6 = c'_2$ $c_2 = 6_{(16)}$ $i=3 : t_3 = -1$ $A_{(16)} + (-1)_{(16)} > 0_{(16)}$ $A_{(16)} + (-1)_{(16)} - 0_{(16)} = 10 - 1 - 0 = 9 = c'_3$ $c_3 = 9_{(16)}$ **Example 2:** $20053_{(6)} - 4444_{(6)} = ?_{(6)}$ $-1 \ -1 \ 0 \ -1 \ 0$ $m=4, n=3, t_0 = 0$ $20053_{(6)} - i=0 :$ $04444_{(6)} \quad 3_{(6)} + 0_{(6)} < 4_{(6)} \Rightarrow \text{borrow one unit from position } i+1=1, \text{ which becomes } 10_{(6)} \text{ in the current position and } t_1 = -1$ $11205_{(6)} \quad 3_{(6)} + 0_{(6)} + 10_{(6)} - 4_{(6)} = (\text{conversion in decimal}) = 3 + 6 - 4 = 5 = c'_0$ $c_0 = 5_{(6)}$ $i=1 : t_1 = -1$ $5_{(6)} + (-1)_{(6)} \geq 4_{(6)} \Rightarrow t_2 = 0$ $5_{(6)} + (-1)_{(6)} - 4_{(6)} = 5 - 1 - 4 = 0 = c'_1$ $c_1 = 0_{(6)}$ $i=2 : t_2 = 0$ $0_{(6)} + 0_{(6)} < 4_{(6)} \Rightarrow \text{borrow} \Rightarrow t_3 = -1$ $0_{(6)} + 0_{(6)} + 10_{(6)} - 4_{(6)} = 6 - 4 = 2 = c'_2$ $c_2 = 2_{(6)}$ $i=3 : t_3 = -1$ $0_{(6)} + (-1)_{(6)} < 4_{(6)} \Rightarrow \text{borrow} \Rightarrow t_4 = -1$ $0_{(6)} + (-1)_{(6)} + 10_{(6)} - 4_{(6)} = -1 + 6 - 4 = 1 = c'_3$ $c_3 = 1_{(6)}$ $i=4 : t_4 = -1$ $2_{(6)} + (-1)_{(6)} > 0_{(6)}$ $2_{(6)} + (-1)_{(6)} - 0_{(6)} = 2 - 1 = 1 = c'_4$ $c_4 = 1_{(6)}$

Example 3: $110001011000_{(2)} - 1110110011_{(2)} = ?_{(2)}$

0 -1 -1 -1 0 -1 0 0 -1 -1 -1 0

$110001011000_{(2)} -$
 $001110110011_{(2)}$

$100010100101_{(2)}$